SAMBAR: Selective Anchoring via Method of Multipliers for Balanced Knowledge Acquisition and Retention in Vision-Language-Action Models

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Motivation

Imagine your robot assistant already knows how to load a dishwasher and fold laundry.

Now you want it to learn to clean an aquarium.

It learns to clean the aquarium but forgets how to load the dishwasher.

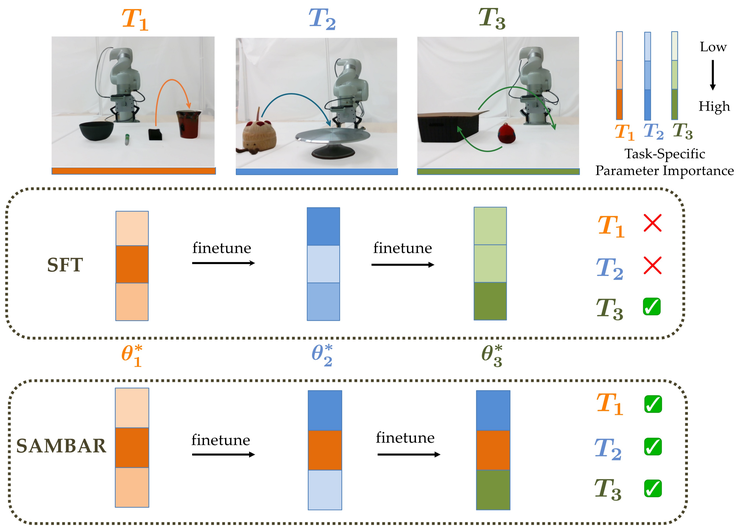


Can we develop a replay-free continual learning method while balancing knowledge acquisition and retention?

Our Contributions

We propose SAMBAR, a replay-free continual learning algorithm for finetuning Vision-language-Action (VLA) models.

- SAMBAR casts continual learning as a constrained optimization problem and solves it with the method of multipliers [1].
- SAMBAR selectively anchors the parameters critical to previous tasks to preserve past knowledge, leaving other parameters free for new task acquisition.

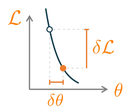


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Computing Parameter's Importance



A parameter is more important to a task when a small change in its value causes a large change in the task loss.



We estimate the importance matrix as

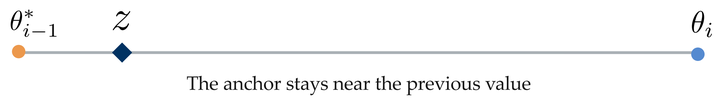
$$\Omega_i = \operatorname{diag}\left(\frac{1}{N_B}\sum_{j=1}^{N_B} \nabla_{\theta_i} L^{BC}(\theta_i; b_j)\left(\nabla_{\theta_i} L^{BC}(\theta_i; b_j)\right)'\right),$$

where b_j is a sample from the batch and N_B is the batch size.

Importance-weighted Anchor

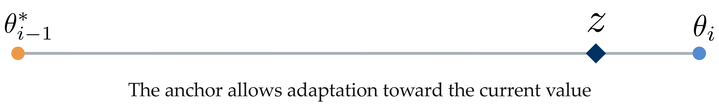
Previous Task Parameters — **Current Parameters**

Important to earlier tasks



The anchor stays near the previous value

Important to the new task



The anchor allows adaptation toward the current value

We then use this parameter anchor z given below in the optimization.

$$z^k = \left(\Omega_i + \bar{\Omega}_{i-1}\right)^{-1}\left(\Omega_i\theta_i^k + \bar{\Omega}_{i-1}\theta_{i-1}^*\right).$$

SAMBAR

Method of multipliers [1] with Importance Weighted Anchor:

At an optimization step k,

Primal Update: $\theta_i^k = \arg\min_{\theta_i} L^{BC}(\theta_i; \mathcal{D}_i) + (\theta_i - z^{k-1})' u^{k-1} + \frac{\rho}{2}\|\theta_i - z^{k-1}\|^2$

Dual Update: $u^k = u^{k-1} + \rho\left(\theta_i^k - z^{k-1}\right)$

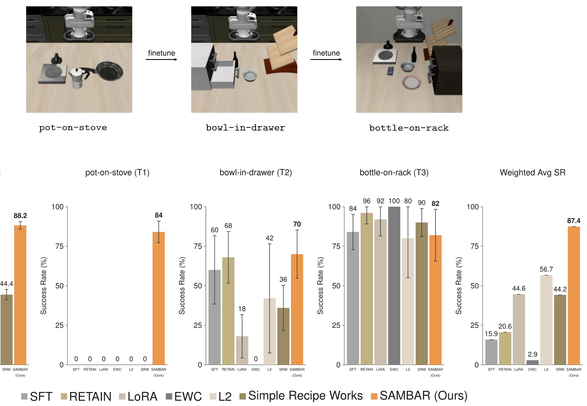
Anchor Update: $z^k = \left(\Omega_i + \bar{\Omega}_{i-1}\right)^{-1}\left(\Omega_i\theta_i^k + \bar{\Omega}_{i-1}\theta_{i-1}^*\right).$

u : dual variable, ρ : hyperparameter for penalty strength, Ω_i : importance matrix

θ_{i-1}^* : optimal parameters that perform well on all prior tasks.

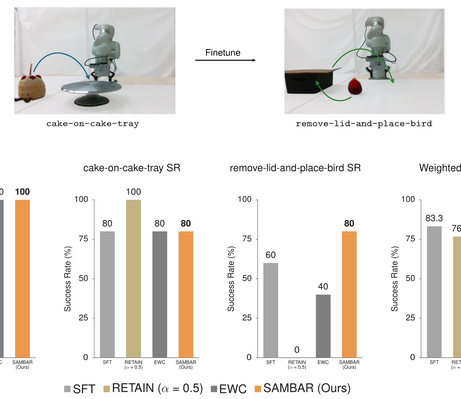
θ_i : parameters for the current task being optimized.

Sequential Finetuning in LIBERO



SAMBAR is the only method which remembers the first finetuning task.

Sequential Finetuning in Hardware



SAMBAR has 100% success on four pretraining tasks and achieves 80% on each of two sequentially learned tasks on hardware.

Conclusion

- SAMBAR balances knowledge acquisition and retention using importance estimates.
- After finetuning on three tasks in sequence, every baseline has fallen to 0.0% on the first task it was taught, while SAMBAR still performs it 84.0% of the time.
- SAMBAR has 100% success on four pretraining tasks and achieves 80% on each of two sequentially learned tasks on hardware.

Website

References

Full text available on ArXiv at https://arxiv.org/abs/2609.32108

[1] S. Boyd et al., "Distributed optimization and statistical learning via the alternating direction method of multipliers," Foundations and Trends in Machine Learning, vol. 3, 2011.